

Inverse Problems in Geophysics

Part 2: Matrix problems and Least Squares

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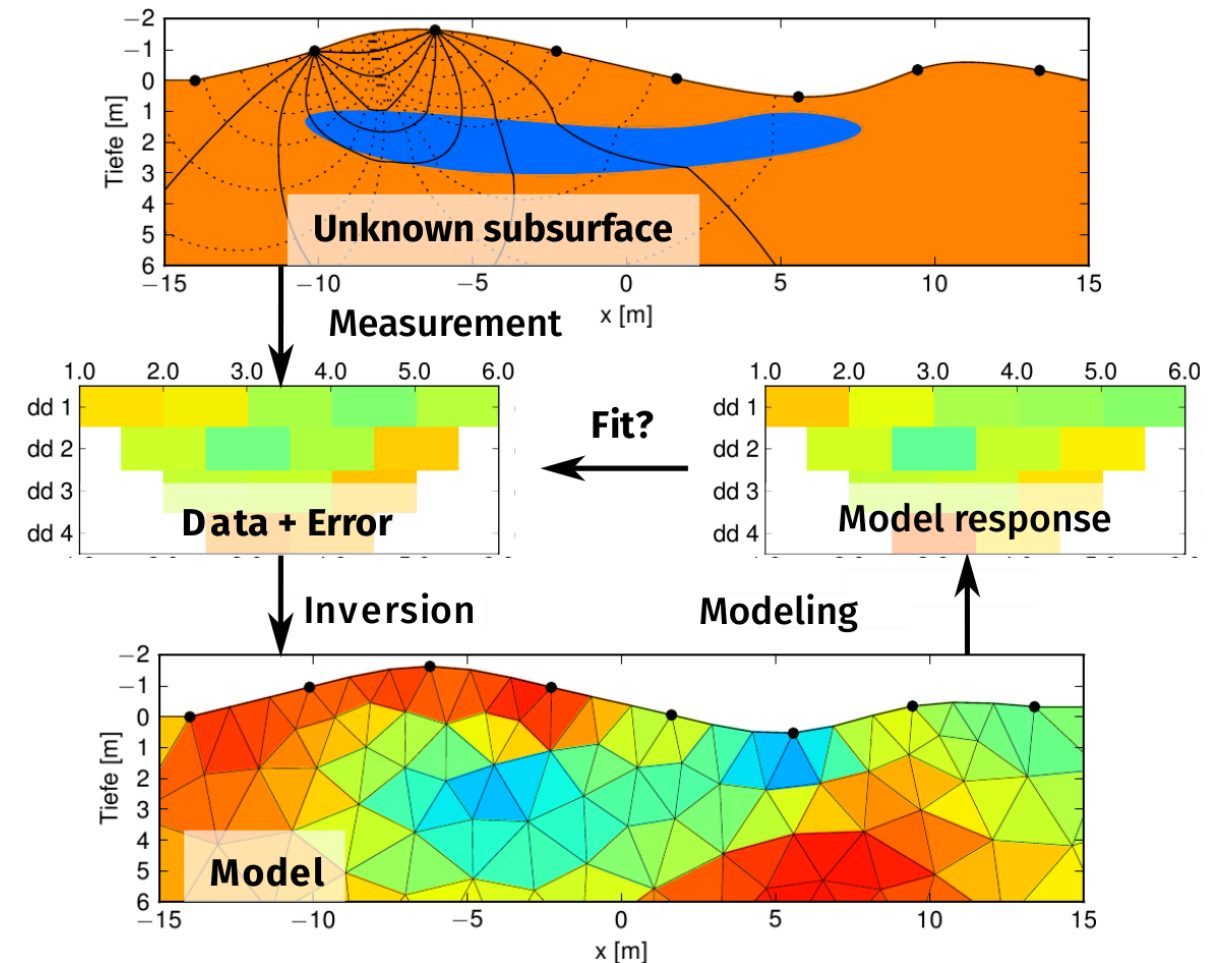
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Recap

- Inversion: create model $\mathbf{m}$ that fits the data $\mathbf{d}$
- data vector along with measuring error
- model vector describes the subsurface
- number of data and model parameters important

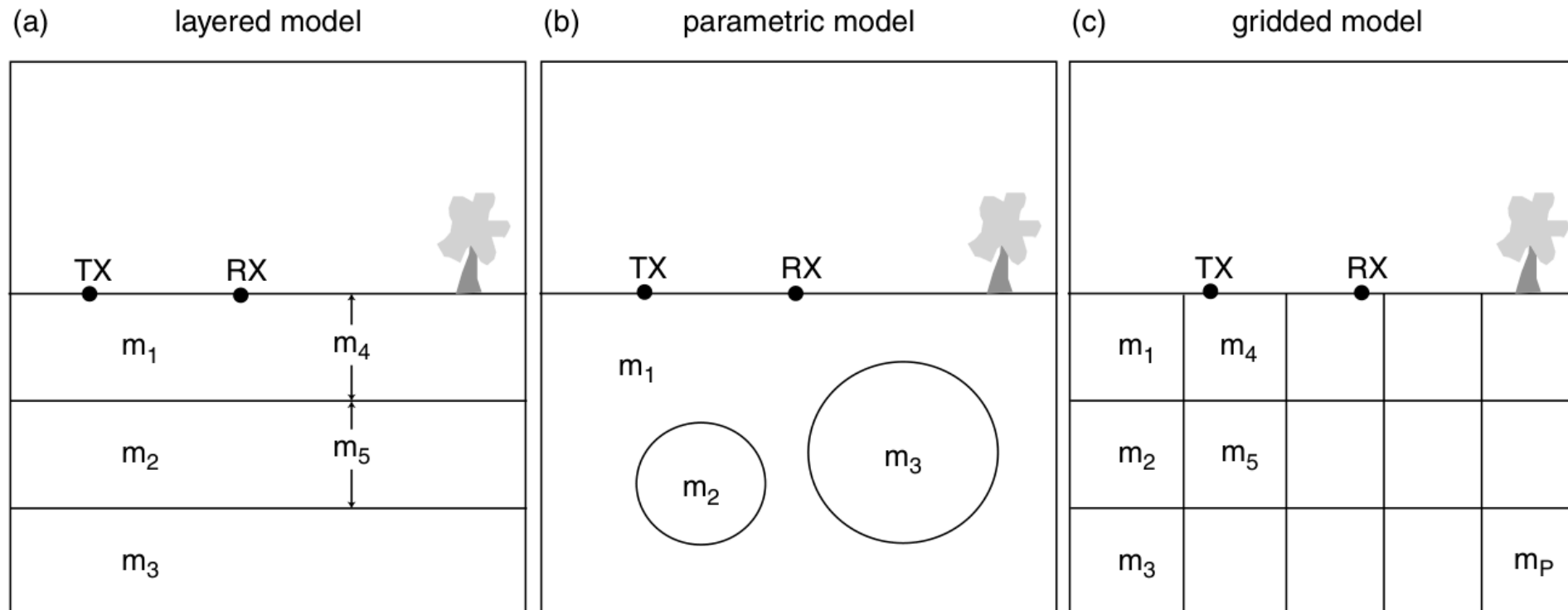
Geophysical Workflow

1. Data acquisition
2. Preprocessing (QC, filtering)
3. Parameterization (mesh)
- 4. Inversion**
5. Fit of data & model response
6. Postprocessing & visualization
7. Interpretation



Model

- numerical parameterization of subsurface (our assumption)



Model types (Everett, 2015)

Model types

subsurface described by *model vector* $\mathbf{m} = [m_1, m_2, \dots, m_M]^T$

Independent parameters:

- seismology: earthquake location, stress, principal axis
- gravity: position and depth of anomaly, size, density contrast
- spectroscopy (e.g. SIP): function parameters (e.g. Cole-Cole)
- layered model thicknesses and values (ρ , v)

Model types

subsurface described by *model vector* $\mathbf{m} = [m_1, m_2, \dots, m_M]^T$

Discrete functions of space, time, frequency

(same property aligned along axes)

- refraction: depth (and velocity) of refractor
- distribution of a parameter in space (and time)
- spectroscopy: Fourier or Debye distribution
- e.g. 2D/3D resistivity/velocity/density distribution

Inversion

i Inverse problem: determine $\mathbf{m}$ so that data are explained

$$\mathbf{d} \approx \mathbf{f}(\mathbf{m})$$

i Why not

$$\mathbf{d} = \mathbf{f}(\mathbf{m})$$

Because $\mathbf{d}$ contains random noise not to be explained.

Minimization problem

i The (data) objective function

$$\Phi_d = \|\mathbf{d} - f(\mathbf{m})\|_2^2 = \sum_i^N (d_i - f_i(\mathbf{m}))^2 \rightarrow \min$$

Take into account errors: Explain model within error (ϵ) bounds

$$\Phi_d = \left\| \frac{\mathbf{d} - f(\mathbf{m})}{\boldsymbol{\epsilon}} \right\|_2^2 = \sum_i^N \left(\frac{d_i - f_i(\mathbf{m})}{\epsilon_i} \right)^2 \rightarrow \min$$

error model $\boldsymbol{\epsilon} = [\epsilon_1, \epsilon_2, \dots, \epsilon_N]^T$ (assumption of noise standard deviation)

Correctness

Well-posed problems (Hadamard)

- There is a solution,
- it is uniquely defined &
- depends steadily from input data (small variations lead to small model deviations)

Ill-posed problem

- There is no model to fit the data perfectly
- Many models can fit the data within errors
- Small data variations can lead to large model deviations

Occam's razor - a fundamental rule

! novacula Occami

Pluralitas non est ponenda sine neccesitate!

(William of Ockham, Scottish philosopher and theologian, 14th century)

i Principle of Parsimony

Entities must not be multiplied beyond necessity.

Of two competing theories, the simpler explanation is to be preferred.

From all models fitting the data, choose the simplest!

Linear problems

$\mathbf{f}(m)$ is linear with respect to $\mathbf{m} \Rightarrow$ write as matrix-vector product

$$\mathbf{d} = G\mathbf{m} + n$$

Examples

Gravity, Magnetics, Magnetic Resonance, VSP, straight-ray tomography

Linear problems

$\mathbf{f}(m)$ is linear with respect to $\mathbf{m} \Rightarrow$ write as matrix-vector product

$$\mathbf{d} = G\mathbf{m} + n$$

Problem

$\mathbf{m} = G^{-1}\mathbf{d}$? No, because is usually not invertible, not even quadratic.

Types of inverse problems (naive)



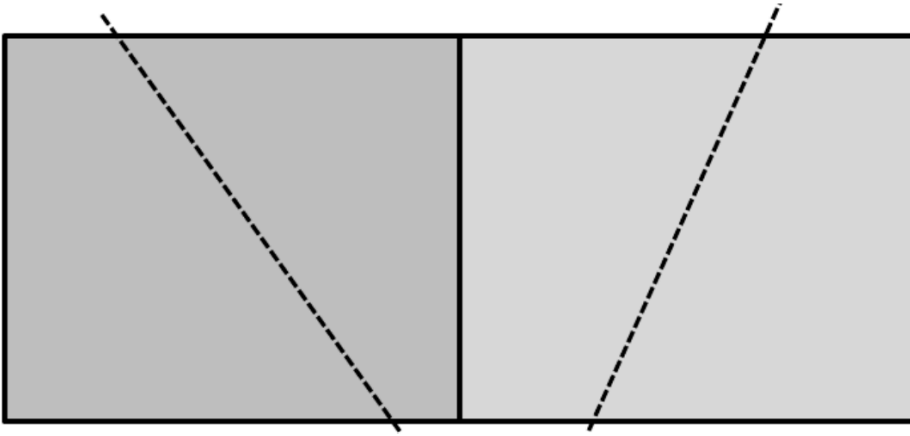
- every row stands for a measurement (data point)
- every column represents a model parameter

$$\mathbf{d} = G\mathbf{m} + n \Rightarrow \mathbf{G} \in \mathfrak{R}^{N \times M}$$

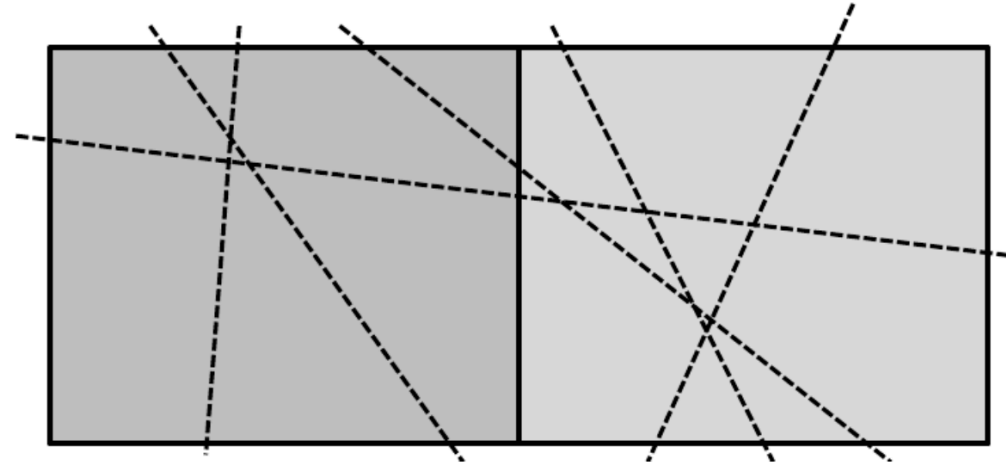
- $N = M$: even-determined
- $N > M$: over-determined problem (no existing solution)
- $N < M$: under-determined problem (no unique solution)
- mostly: (both over- and) under-determined model parts

Types of inverse problems (Menke, 2012)

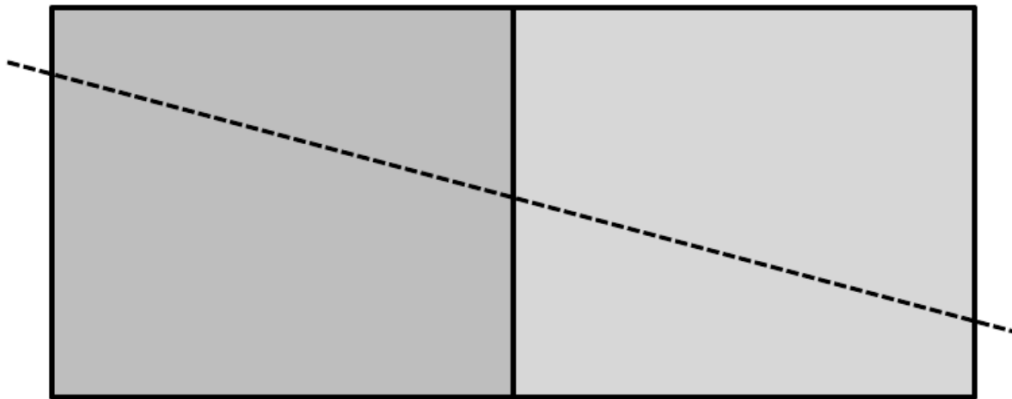
Even-determined



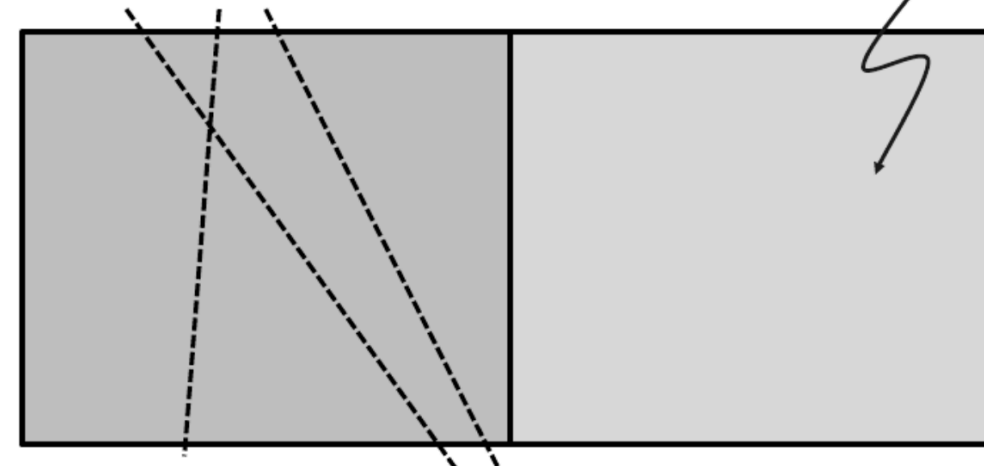
Over-determined



Under-determined



Mixed-determined



A simple matrix problem

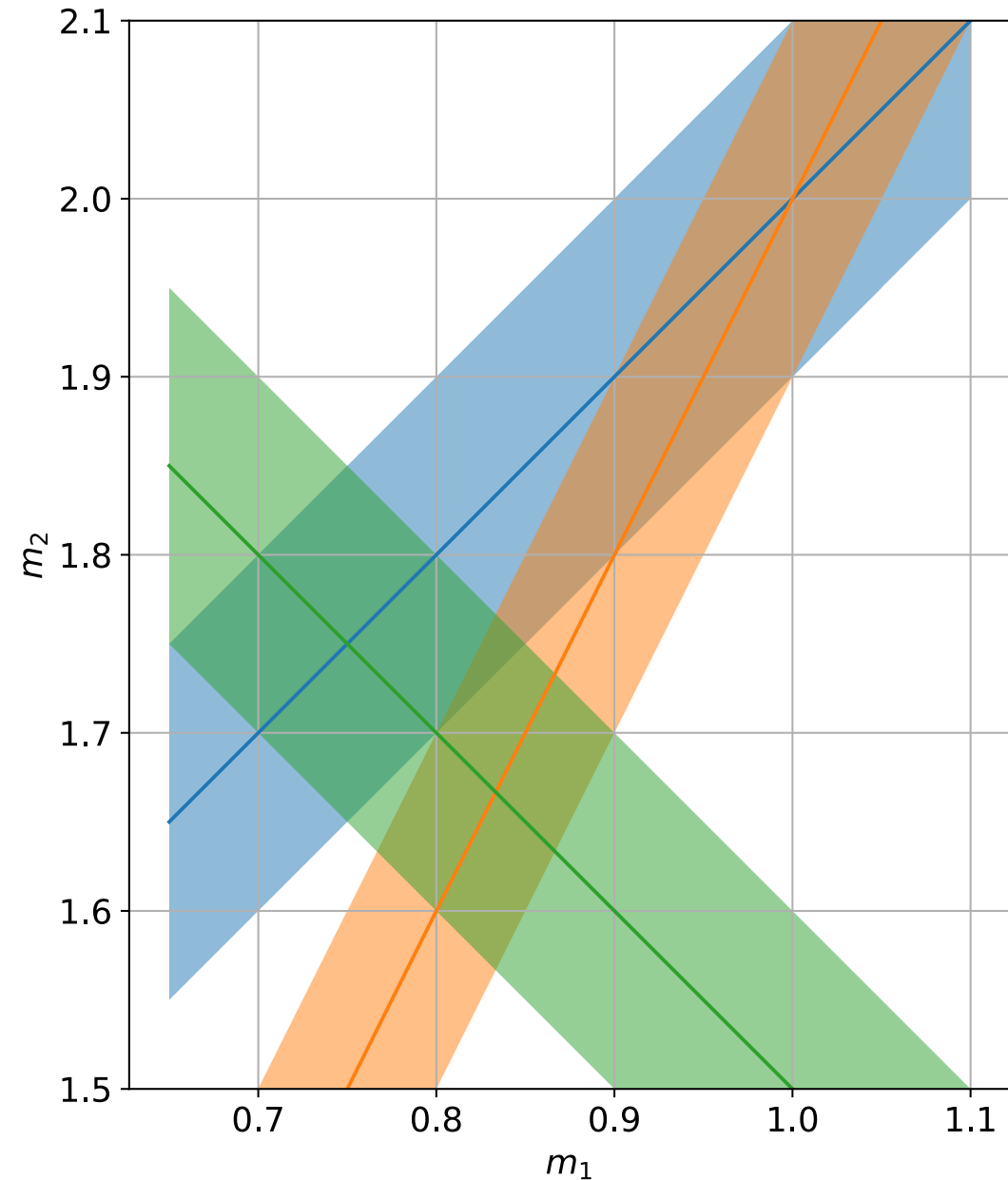
A simple matrix problem with errors

$$\mathbf{G} = \begin{pmatrix} 1 & -1 \\ 2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$m_2 = (d_i - G_{i1} \cdot m_1) / G_{i2}$$

$$\delta m_2 = \delta d_i / G_{i2}$$

Data error of $\delta d = 0/0.05/0.1$



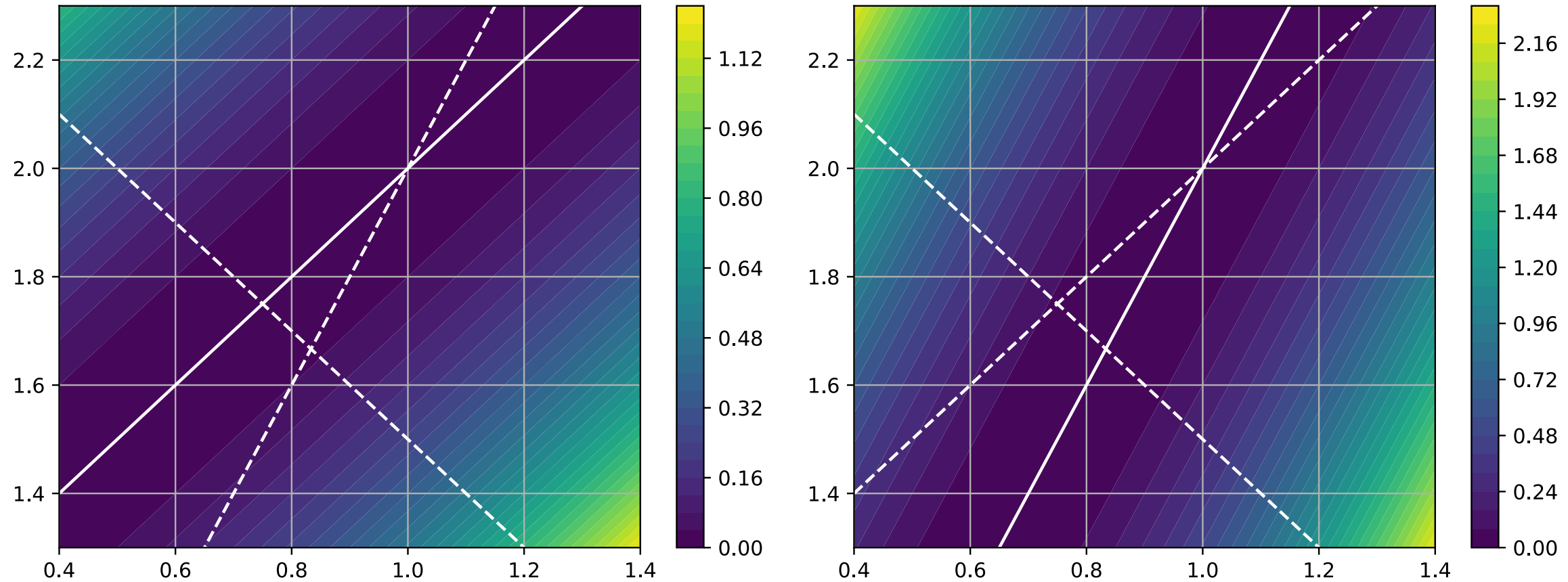
The objective function

We minimize the L2-norm (summed squared distances) of the residual between data and model response:

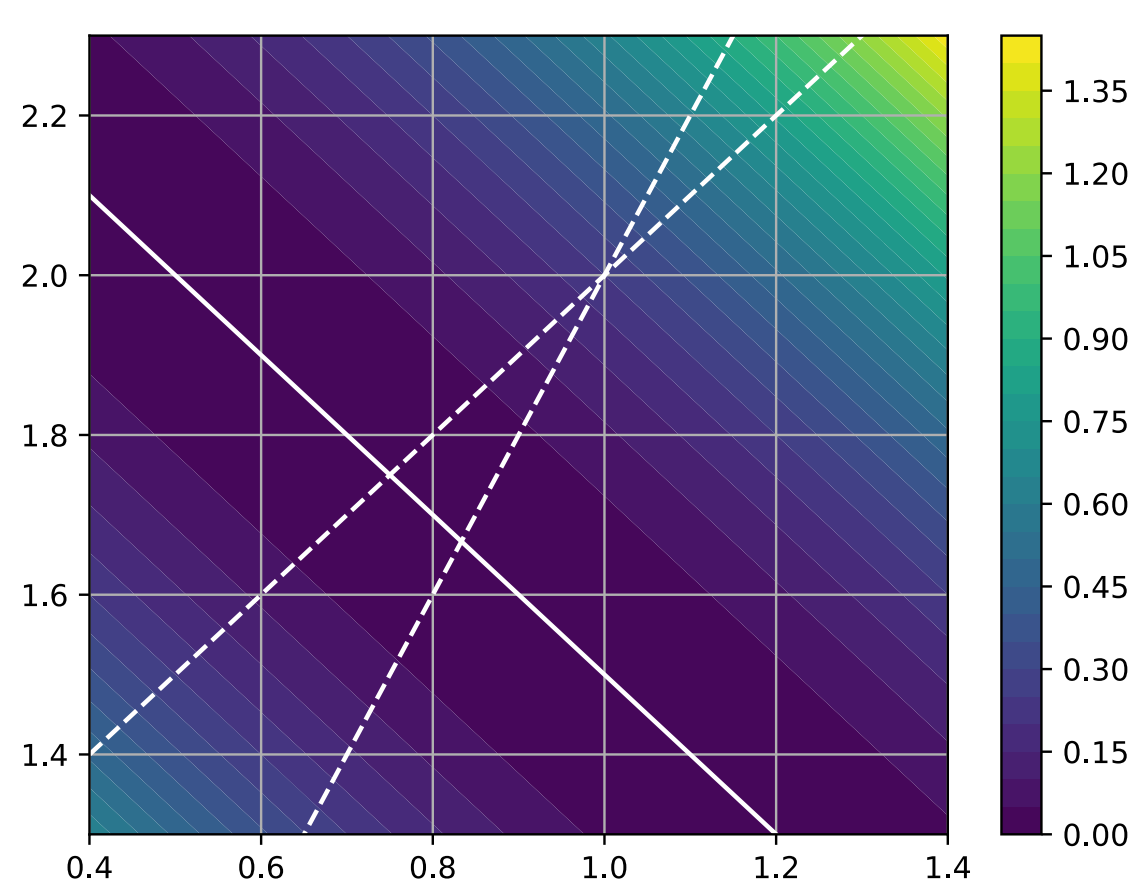
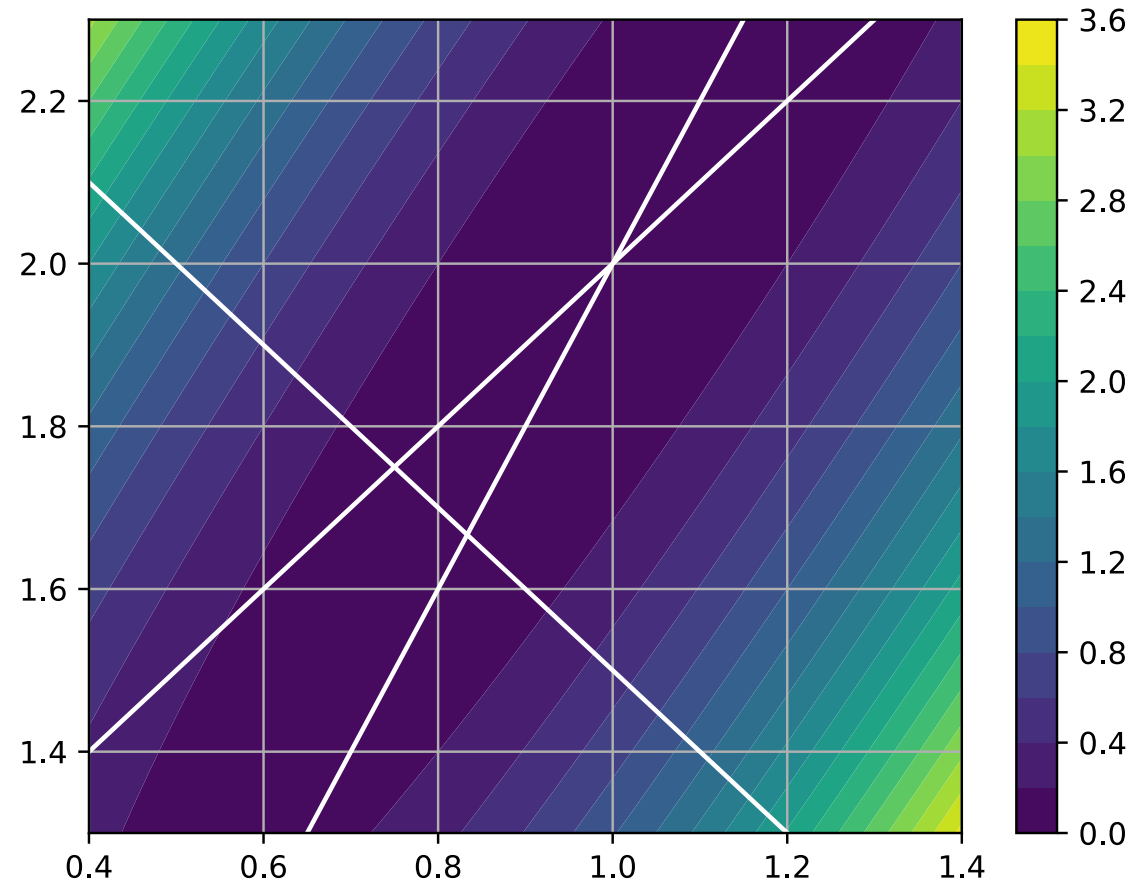
$$\Phi = \|\mathbf{d} - G\mathbf{m}\|_2^2 = \sum_1^N (d_i - g_i(\mathbf{m}))^2 \rightarrow \min$$

Let's compute the objective function for a range of values for m_1 and m_2 (grid search).

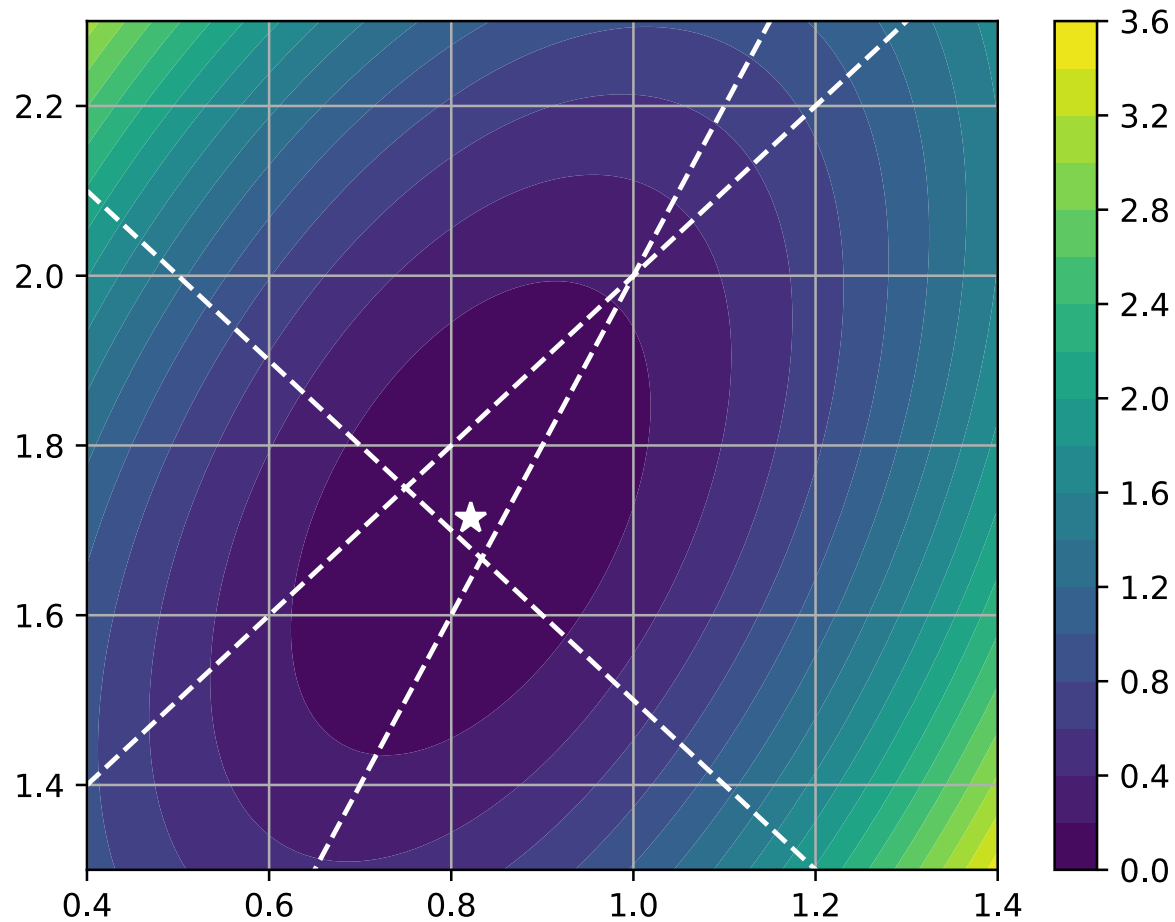
Distance function for the first two data



Distance function of 1st + 2nd and 3rd



Total objective function



- Add all three squared distances.
- ellipsoidal minimum shape
- The minimum pixel (grid spacing of 0.01) lies at (0.82, 1.71).

How can we determine the absolute minimum value?

The method of least squares

Derivation

$$\Phi = \|\mathbf{d} - \mathbf{G}\mathbf{m}\|_2^2 = (\mathbf{d} - \mathbf{G}\mathbf{m})^T (\mathbf{d} - \mathbf{G}\mathbf{m})$$

$$\nabla_m = \left(\frac{\partial}{\partial m_1}, \frac{\partial}{\partial m_2}, \dots, \frac{\partial}{\partial m_M} \right)^T$$

$$\nabla_m \Phi = \nabla_m (\mathbf{G}\mathbf{m} - \mathbf{d})^T (\mathbf{G}\mathbf{m} - \mathbf{d}) = 0$$

$$\nabla_m \Phi = \nabla_m (\mathbf{m}^T \mathbf{G}^T - \mathbf{d}^T) (\mathbf{G}\mathbf{m} - \mathbf{d}) = 0$$

$$\nabla_m \Phi = \nabla_m \mathbf{m}^T \mathbf{G}^T (\mathbf{G}\mathbf{m} - \mathbf{d}) = 0$$

Derivation (cont.)

$$\nabla_m \Phi = \nabla_m \mathbf{m}^T \mathbf{G}^T (\mathbf{G} \mathbf{m} - \mathbf{d}) = 0$$

$$\nabla_m \Phi = \mathbf{G}^T (\mathbf{G} \mathbf{m} - \mathbf{d}) = \mathbf{G}^T \mathbf{G} \mathbf{m} - \mathbf{G}^T \mathbf{d} = 0$$

$$\Rightarrow \quad \mathbf{m} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \mathbf{d} = \mathbf{G}^\dagger \mathbf{d}$$

$$\text{mit } \mathbf{G}^\dagger = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T$$

(generalized inverse or Pseudo-inverse `pinv`, in Matlab/Julia `G\d`)

The solution $\mathbf{m}_{LS} = \mathbf{G}^\dagger \mathbf{d}$ is the **least-squares solution**

Application to the matrix problem

Resolution analysis

- How does reality propagate into our image?
- What is the quality of the inversion result?
- What is the contribution of the individual data?

Model resolution

$$\mathbf{d} = \mathbf{G}\mathbf{m}^{\text{true}} + \mathbf{n}$$

Matrix inversion with inverse operator $\mathbf{G}^\dagger$:

$$\mathbf{m}^{\text{est}} = \mathbf{G}^\dagger \mathbf{d} = \mathbf{G}^\dagger \mathbf{G} \mathbf{m}^{\text{true}} + \mathbf{G}^\dagger \mathbf{n} = \mathbf{R}^M \mathbf{m}^{\text{true}} + \mathbf{G}^\dagger \mathbf{n}$$

with the **model resolution matrix** $\mathbf{R}^M = \mathbf{G}^\dagger \mathbf{G}$

$\Rightarrow$ How is the true model ($\mathbf{m}^{\text{true}}$) reflected in the estimated ($\mathbf{m}^{\text{est}}$)?

Diagonal of $\mathbf{R}^M$: resolution of individual parameters (0-1)

Data resolution

$$\mathbf{m}^{\text{est}} = \mathbf{G}^\dagger \mathbf{d}^{\text{obs}}$$

How are the data explained by the model?

$$\mathbf{d}^{\text{est}} = \mathbf{G}\mathbf{m}^{\text{est}} = \mathbf{G}\mathbf{G}^\dagger \mathbf{d}^{\text{obs}} = \mathbf{R}^D \mathbf{d}^{\text{obs}}$$

with the data resolution (information density) matrix:

$$\mathbf{R}^D = \mathbf{G}\mathbf{G}^\dagger$$

Diagonal of $\mathbf{R}^D$: information content of individual data

Overdetermined problems

$$\mathbf{G}^\dagger = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T$$

$$\Rightarrow \mathbf{R}^M = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \mathbf{G} = \mathbf{I}$$

perfect model resolution

$$\mathbf{G}^\dagger = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \quad \Rightarrow \quad \mathbf{R}^D = \mathbf{G}(\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T$$

Wrap-up overdetermined problems ($N > M$)

- objective function Φ as squared data misfit
- weighting of individual data (unitless, more flexible & objective)
- broad minimum of the objective function
- grid search to plot $\phi \Rightarrow$ only nice for $M=2$
- least squares method yields
- resolution matrices for model (perfect) and data (distributed)

Appendix

The generalized inverse

The matrix

$$\mathbf{G}^\dagger = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T$$

is called pseudo-inverse (Moore-Penrose inverse)

The solution $\mathbf{m}_{LS} = \mathbf{G}^\dagger \mathbf{d}$ is the least-squares solution

Error weighting

Unweighted residual norm (root-mean square RMS)

$$\|\mathbf{d} - \mathbf{f}(\mathbf{m})\| = \sqrt{1/N \sum (d_i - f_i(\mathbf{m}))^2}$$

Weighting by individual error ϵ_i (chi-square value):

$$\chi^2 = \frac{1}{N} \sum \left(\frac{d_i - f_i(\mathbf{m})}{\epsilon_i} \right)^2 \rightarrow \min$$

In case of exact error estimates: $\chi^2 = 1$

Error weighting

replace d_i by $\hat{d}_i = d_i/\epsilon_i$ leads to

$$\mathbf{m} = (\hat{\mathbf{G}}^T \hat{\mathbf{G}})^{-1} \hat{\mathbf{G}}^T \hat{\mathbf{d}}$$

with $\hat{\mathbf{G}} = \text{diag}(1/\epsilon_i) \cdot \mathbf{G}$

Derivation (1)

$$\Phi = \|\mathbf{d} - \mathbf{G}\mathbf{m}\|_2^2 = (\mathbf{d} - \mathbf{G}\mathbf{m})^T (\mathbf{d} - \mathbf{G}\mathbf{m})$$

$$\Phi = (\mathbf{G}\mathbf{m} - \mathbf{d})^T (\mathbf{G}\mathbf{m} - \mathbf{d})$$

$$\frac{\partial \Phi}{\partial \mathbf{m}} = \frac{\partial}{\partial \mathbf{m}} (\mathbf{G}\mathbf{m} - \mathbf{d})^T (\mathbf{G}\mathbf{m} - \mathbf{d}) + (\mathbf{G}\mathbf{m} - \mathbf{d})^T \frac{\partial}{\partial \mathbf{m}} (\mathbf{G}\mathbf{m} - \mathbf{d}) =$$

$$\mathbf{G}^T \mathbf{G}\mathbf{m} - \mathbf{G}^T \mathbf{d} + \mathbf{G}^T \mathbf{G}\mathbf{m} - \mathbf{G}^T \mathbf{d} = 0$$

$$\mathbf{G}^T \mathbf{G}\mathbf{m} = \mathbf{G}^T \mathbf{d} \quad \Rightarrow \quad \mathbf{m} = \mathbf{G}^\dagger \mathbf{d} \quad \text{with} \quad \mathbf{G}^\dagger = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T$$

Derivation (2)

$$\Phi = (\mathbf{d} - \mathbf{G}\mathbf{m})^T (\mathbf{d} - \mathbf{G}\mathbf{m}) = \sum_i \left[(d_i - \sum_j G_{ij}m_j)(d_i - \sum_k G_{ik}m_k) \right]$$

$$\Phi = \sum_i \left[d_i d_i - d_i \sum_k G_{ik}m_k - d_i \sum_j G_{ij}m_j + \sum_j G_{ij}m_j \sum_k G_{ik}m_k \right]$$

$$\Phi = \sum_i d_i d_i - 2 \sum_j m_j \sum_i d_i G_{ij} + \sum_i \sum_j \sum_k m_j G_{ij} G_{ik} m_j m_k$$

$$\Phi = \sum_i d_i d_i - 2 \sum_j m_j \sum_i d_i G_{ij} + \sum_j \sum_k m_j m_k \sum_i G_{ij} G_{ik}$$

$$\partial \Phi = \partial m_q = \sum_i \sum_k (\delta_{iq} m_k + m_j \delta_{ik}) \sum_i G_{ij} G_{ik} - 2 \sum_j \delta_{iq} \sum_i G_{ij} d_i$$

$$0 = 2 \sum_k \sum_i G_{iq} G_{ik} - 2 \sum_i G_{iq} d_i = 2 \mathbf{G}^T \mathbf{G} - 2 \mathbf{G}^T \mathbf{d}$$

Derivation by change $t\delta\mathbf{m}$

$$\Phi(t) = (\mathbf{d} - \mathbf{G}(\mathbf{m} + t\delta\mathbf{m}))^T (\mathbf{d} - \mathbf{G}(\mathbf{m} + t\delta\mathbf{m}))$$

$$\Phi(t) = (\mathbf{m} + t\delta\mathbf{m})^T \mathbf{G}^T \mathbf{G} (\mathbf{m} + t\delta\mathbf{m}) - 2(\mathbf{m} + t\delta\mathbf{m})^T \mathbf{G}^T \mathbf{d} + \mathbf{d}^T \mathbf{d}$$

$$\Phi(t) = t^2(\delta\mathbf{m}^T \mathbf{G}^T \mathbf{G} \delta\mathbf{m}) + 2t(\delta\mathbf{m}^T \mathbf{G}^T \mathbf{G} \mathbf{m} - \delta\mathbf{m}^T \mathbf{G}^T \mathbf{d}) + (\mathbf{m}^T \mathbf{G}^T \mathbf{G} \mathbf{m} - \mathbf{m}^T \mathbf{G}^T \mathbf{d} + \mathbf{d}^T \mathbf{d})$$

$\Phi(t)$ has a minimum at $t = 0$, therefore $\partial\Phi/\partial t$ must be 0

$$\partial\Phi(t = 0)/\partial t = 2(\delta\mathbf{m}^T \mathbf{G}^T \mathbf{G} \mathbf{m} - \delta\mathbf{m}^T \mathbf{G}^T \mathbf{d}) = 2\delta\mathbf{m}^T (\mathbf{G}^T \mathbf{G} \mathbf{m} - \mathbf{G}^T \mathbf{d})$$

As this holds for every $\delta\mathbf{m}$, we obtain $\mathbf{G}^T \mathbf{G} \mathbf{m} = \mathbf{G}^T \mathbf{d}$